

ASSESSMENT AND INTERNAL VERIFICATION FRONT SHEET (Individual Criteria)

(Note: This version is to be used for an assignment brief issued to students via Classter)

Course Title	B.Sc. (Hons.) in Applied Data Sciences B.Sc. (Hons.) Computer Systems & Networking B.Sc. (Hons.) Creative Computing B.Sc. (Hons.) Cybersecurity B.Sc. (Hons.) Multimedia Software Development B.Sc. (Hons.) Software Development			Lecturer Name & Surname	Antonio Borg Clifford De Raffaele Frankie Inguanez Darren Scerri Elaine Vassallo
Unit Number & Title	ITRSH-506-2101 – Research Design 1				
Assignment Number, Title / Type	01, Research Project / Home				
Date Set	18/02/2026	Deadline Date	31/05/2026		
Student Name		ID Number		Class / Group	

Assessment Criteria	Maximum Mark
KU1.1 Present gathered research in relation to a theme that motivates research in an area/topic.	5
SE1.2 Formulate a research hypothesis, supported with research questions from which research methods will be derived.	10
AA1.3 Illustrate a research pipeline that will be followed to address own research hypothesis.	7
KU1.4 Clarify which research methods will be used to address the identified research questions and how these will be evaluated.	5
AA1.5 Apply every stage of the research pipeline to own research endeavour.	7
KU2.1 Outline the current state of the art of own research topic (h/w, data, alg).	5
KU2.2 Describe different existing sources of data with key features and uses in different research.	5
AA2.3 Contrast own research findings with that of current state of the art.	7
KU3.1 Defend the proposed pipeline by citing own results and that of the current state of the art.	5
SE3.2 Evaluate the outcome of each identified research question.	10
AA3.3 Investigate the extent to which results confirm original hypothesis.	7
AA3.4 Investigate areas in which results suggest the original hypothesis needs modification.	7
KU4.1 Record the work in a report following a proper referencing style.	5
KU4.2 Arrange own work in a scientifically structured manner with proper internal referencing, sectioning and labelling.	5
SE4.3 Criticize own research to propose new hypothesis and research questions for future work.	10
Total Mark	100

Notes to Students:

- This assignment brief has been approved and released by the Internal Verifier through Classter.
- Assessment marks and feedback by the lecturer will be available online via Classter ([Http://mcast.classter.com](http://mcast.classter.com)) following release by the Internal Verifier
- Students submitting their assignment on Moodle/Turnitin will be requested to confirm online the following statements:

Student's declaration prior to handing-in of assignment

- ❖ I certify that the work submitted for this assignment is my own and that I have read and understood the respective Plagiarism Policy

Student's declaration on assessment special arrangements

- ❖ I certify that adequate support was given to me during the assignment through the Institute and/or the Inclusive Education Unit.
- ❖ I declare that I refused the special support offered by the Institute.

Overview

This unit aims to aid you in formulating an understanding of a research project following a scientific research methodology. A staggered assessment approach shall be undertaken, so that you can build your project gradually. Regular communication with your lecturer is key. Following is a schedule and outline of the milestones and respective deliverables:

Schedule

Milestone	Deliverable	Marks
01	Research Topic Identification	37%
02	Final prototype, paper, VIVA presentation	53%
03	VIVA presentation delivery	10%

Milestone 01: Research Topic Formulation

You are to follow the proposed framework to aid in the research topic formulation. For this milestone you are to submit the following:

1. **Title:** Propose a title for your research topic.
2. **Description:** A 100-word description of what the research is all about.
3. **Keywords:** 5 keywords that relate to your research.
4. **Motivation:** A 200-word description on what is the motivation for the chosen topic.
5. **Academic Interest:** A 1,000-word literature review based on 3 published academic papers about the research topic. This literature review should outline what algorithms and datasets are used by academics researching your chosen topic and what results are achieved in recent years. It is recommended that one paper is a systematic literature review about the topic, with two additional reference papers. These papers utilise data and algorithms that can be replicated and that you will use to compare your future results with.
6. **Data Availability:** A table outlining the different datasets used by other academic researchers on the subject matter. The table should include the title, whether they are public/private, the format (text, image, video, audio, etc.), instances, columns, and other topic specific features.
7. **Reference Solutions:** A table outlining sample solutions or tutorials or courses that you shall be basing your research on by replicating then customising.
8. **Resources:** A table outlining the hardware and software utilised/needed to replicate the identified solutions together with benchmark data related to the execution of the solution (execution time, evaluation score, observations, etc.)
9. **Direction:** A 200-word description, based on the identified papers, of key recommendations, suggested hypotheses or research questions outlining the direction that academic or industry is heading in the subject area.
10. **Focus:** 1-2 research questions outlining your research focus.
11. **Evaluation:** For each research question propose a research method on how you will gather data to answer the proposed research question.

Milestone 02: Final prototype, paper, viva presentation

1. The source code for the final working prototype.
2. The dataset(s), if relevant, used for the experimentation.
3. A 3–5-page paper using IEEE conference template (Word or LaTeX) in PDF form. The paper should have the following:
 - a. Title, with author and affiliation details.
 - b. A 150-200 word abstract that outlines the aim of the research, the research methods adopted, and results achieved.
 - c. 5 keywords that define your research.
 - d. An introduction that outlines the area of study, motivation, and importance of this research. It is recommended to include the research aim, hypothesis, and question(s) here.
 - e. A literature review based on the identified academic papers outlining the current state-of-the-art, existing datasets, results by academics using specific solution algorithms.
 - f. A research methodology in which the research pipeline, research methods are outlined.
 - g. A findings section in which you outline the results of your experiment/research. Compare your research with that of other researchers. Evaluate the research questions.
 - h. A conclusion section that is a critical evaluation of the undertaken research (what was achieved, what were the limitations, what is the recommended way forward).
 - i. References in accordance with the IEEE template.
 - j. The VIVA presentation document to be used when presenting your work to your respective lecturer. The presentation is to last for a maximum of 10 minutes and should include a recording (up to 2-minutes) of your working prototype.

Milestone 03: VIVA presentation delivery

1. Deliver the presentation to your lecturer.

Grading Criteria

Milestone 01

Criteria / Task	Low Performance	Average Performance	Best Performance
SE1.2	1-3	4-6	7-10
Identified research	Proposed research is broad, lacking direction or context.	Proposed research identifies key technologies and context yet is lacking depth of thought.	Proposed research is detailed, relevant to specific context and based on good background research.
Research hypothesis	Hypothesis is generic and lacking rigor.	Hypothesis specifies research direction and correlates well with research questions.	Hypothesis is well defined and encapsulates the research direction well.
Research questions	One or two research questions provided that are loosely linked to research hypothesis or are too generic.	Three research questions provided but are lacking depth of thought.	Three research questions provided, contextualised to research, and showing depth of thought.

Criteria / Task	Low Performance	Average Performance	Best Performance
KU1.1	1-2	3	4-5
Reviewed papers	1-2 papers referenced are related to the research topic.	3-4 papers referenced are related to the research topic.	5+ recent papers referenced are related to the research topic.
Depth of review	Review is shallow, giving generic information and lacking flow.	Papers are reviewed giving a considerable overview of the research undertaking.	Reviewed papers are represented wholistically and contribute well to this research.

Criteria / Task	Low Performance	Average Performance	Best Performance
KU2.2	1-2	3	4-5
Review of data sets / resources	A listing of existing assets/datasets with limited information such as author, creation year and basic features given.	Review of assets/datasets or resources used in research are explained in some detail.	Assets/Datasets or resources are reviewed in detail highlighting opinions by other researchers which aids in the defence of choosing a specific dataset or in the decisions taken to create a custom dataset.

Criteria / Task	Low Performance	Average Performance	Best Performance
KU2.1	1-2	3	4-5
Review of state-of-the-art	A listing of existing approaches/solutions with limited information such as author, creation year and basic features given.	Review of different approaches/solutions used in research are explained in some detail.	Solutions or approaches are reviewed in detail highlighting opinions by other researchers which aids in the defence of choosing a solution/approach or in the decisions taken.

Criteria / Task	Low Performance	Average Performance	Best Performance
KU1.4	1-2	3	4-5
Proposed research methods	Research method provided with limited explanation of data gathering or experimentation.	Research method provided with detailed approach for data gathering and experimentation.	Excellent choice of research method(s) proposed, aligning with other research. Use of existing resources and creation of own, with experimentation that allows for comparison with third party.

Criteria / Task	Low Performance	Average Performance	Best Performance
AA1.3	1-3	4-5	6-7
Proposed research pipeline/plan	A basic generic pipeline/plan, that does not show inspiration from existing literature or does not show research rigour.	A pipeline/plan that shows rigour and structure, aiding in the undertaking of the research project.	A pipeline/plan inspired from existing literature and contextualised for the research.

Milestone 02

Criteria / Task	Low Performance	Average Performance	Best Performance
KU4.1	1-2	3	4-5
Referencing style	Referencing style used poorly, with substantial missing fields.	Referencing style used properly. Yet some fields are missing such as DOI, incorrect abbreviation of authors, missing URL, or other fields such as accessed date.	Referencing style used properly. Citations are constructed properly with DOI, correct use of corporate author if relevant, accessed date if relevant.

Criteria / Task	Low Performance	Average Performance	Best Performance
KU4.2	1-2	3	4-5
Scientific Paper	Template not used properly, and paper not structured as expected.	Paper follows general template and provides guidelines, yet shortcomings are visible.	Template and submission guidelines followed perfectly.

Criteria / Task	Low Performance	Average Performance	Best Performance
KU3.1	1-2	3	4-5
Executed research pipeline defence in research methodology	A generic pipeline is provided with no justification or reference to other research.	A good pipeline is presented with very limited justification or reference to other research.	A good pipeline is presented with citations for inspiration and comparison.

Criteria / Task	Low Performance	Average Performance	Best Performance
AA1.5	1-2	3-4	5-7
Documentation of work undertaken in the Findings & Results section (ideally following pipeline)	Even though work is documented, it is difficult at what stage of the pipeline it occurred.	Work is documented and referenced along the research pipeline yet is limited in detail.	Work is documented rigorously and referenced well along the research pipeline.
Prototype demonstration	Basic demonstration of working prototype with some errors or flaws.	Complete demonstration of working prototype with some flaws.	Demonstration of working prototype and experimentation.

Criteria / Task	Low Performance	Average Performance	Best Performance
AA2.3	1-2	3-4	5-7
Comparison with other work	Limited to no comparison of own work with third party.	Comparison of work done with third party, yet evaluation does not align perfectly.	A fair comparison of own work with third party using same evaluation technique/metric.

Criteria / Task	Low Performance	Average Performance	Best Performance
SE3.2	1-3	4-6	7-10
Research question evaluation	Research questions are poorly or incorrectly answered.	Research questions are answered yet are biased or lacking rigour.	Research questions are answered with rigour and in a fair manner.

Criteria / Task	Low Performance	Average Performance	Best Performance
AA3.3	1-2	3-4	5-7
Hypothesis evaluation	Hypothesis is poorly evaluated.	Hypothesis is evaluated but contains bias.	Hypothesis is evaluated fairly and rigorously.

Criteria / Task	Low Performance	Average Performance	Best Performance
AA3.4	1-2	3-4	5-7
Alternate Hypothesis proposed	One superficial alternate hypothesis proposed.	Various alternate hypotheses proposed with no or little justification.	Alternate hypotheses proposed evidenced with results from research.

Milestone 03

Criteria / Task	Low Performance	Average Performance	Best Performance
SE4.3	1-3	4-6	7-10
VIVA presentation	A basic presentation of work with limited to no criticism	A good presentation of work, with fair self-criticism, yet superficial future recommendations.	An excellent presentation of work, with identified limitations, recommended new and meaningful hypothesis, research questions and future work.